Nitin Bansal

LinkedIn: linkedin.com/in/nitin-bansal4 Email: nb0350422@gmail.com
GitHub: github.com/Nitinbansal-2004 Mobile: +91-7340459017

SKILLS

- **Languages:** Python, C++(Major), Java, C, HTML, CSS
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Tools/Platforms:** Git, GitHub, XAMPP, Netlify, VS Code, Postman, Excel, Power-Bi
- **Core Fundamentals:** Data Structures & Algorithms (DSA), OOP, DBMS, Algorithm Design & Optimization

PROJECTS

- Ai-Driven CPU Scheduling– Intelligent System Resource Optimizer — Feb 2025 – Apr 2025

  Technologies: Python, HTML, CSS, JavaScript, Chart.js, Node.js, WebSockets, SystemInformation API
  - Developed a **real-time system monitoring and smart CPU scheduling tool** to track live processes, CPU load, memory usage, and disk activity through continuous metric collection.
  - Implemented **an auto-termination mechanism** to detect long-running or resource-intensive processes and safely terminate them, optimizing system performance and reducing energy consumption.
  - Built a **WebSocket-based backend** to stream live system metrics with low latency, enabling real-time updates and responsive system monitoring.
  - Designed an **interactive frontend dashboard** using Chart.js to visualize system performance, display alerts, and provide process-level controls for user-driven optimization.
  - Integrated optional **AI-based prediction modules (Python)** to analyze usage patterns and support energy-efficient (green) computing decisions.
- **AI Driven CPU Scheduling-Intelligent System Resource | College Project** — Jan 2025 – Feb 2025

  Technologies: Python, yFinance, NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, XGBoost, PCA
  - Developed a stock market analysis and prediction tool using live 30-minute interval data from yFinance to analyze short-term market behavior.
  - Performed **statistical analysis and technical indicators** including SMA/EMA trend studies, correlation mapping, and candlestick pattern visualization for market insights.
  - Built and evaluated **machine learning models** such as Linear Regression, Random Forest, and XGBoost to predict stock closing prices with high accuracy.
  - Applied **Principal Component Analysis (PCA)** for dimensionality reduction, enhancing pattern visualization and improving model efficiency.

TRAINING

- **Competitive Programming Certification | AlgoTutor Academy** — Jun 2025 – Jul 2025

  Skills: C++, Data Structures & Algorithms (DSA), STL, Complexity Analysis
  - Solved **100+** competitive programming problems focused on arrays, graphs, dynamic programming, and **time/space optimization.**
  - Practiced **timed coding challenges** using C++ STL, improving solution speed and algorithmic efficiency.
  - Strengthened **debugging, logical reasoning,** and **contest-style coding** under constraints.

CERTIFICATES

- Build Generative AI Apps and Solutions with No-Code Tolls | Coursera — Mar 2025
- Prompt Engineering: Generative AI & LLMs – Infosys Springboard — Aug 2025 – Sep 2025
- Privacy & Security in Online Social Media – NPTEL (Govt. of India) — Jan 2025 – Apr 2025
- Live Python Basics – CSE Pathshala — Feb 2024 – Mar 2024
- Responsive Web Design – FreeCodeCamp(300 hours) — Sep 2023–Dec 2023

ACHIEVEMENTS

- **Competitive Programming & Data Structures**
  - Engineered optimized solutions for **300+ Data Structures & Algorithms (DSA) problems** on **LeetCode** and **GeeksforGeeks,** emphasizing time complexity optimization, space efficiency, and scalable algorithm design.
  - Ranked among the **Top ~4,600 globally** in a **CodeChef** rated contest with **1051+ rating,** demonstrating strong competitive programming, complexity analysis, and fast problem-solving under time constraints.
  - Earned **HackerRank Problem Solving (Intermediate)** Certificate by clearing a **skill-based assessment** covering advanced data structures, algorithms, and logical reasoning.

EDUCATION

- **Lovely Professional University** — Punjab, India

  Bachelor of Technology in Computer Science and Engineering | CGPA: 7.96 — Aug 2023 – Present
- **Govenment Sr. Sec. School** — Bari, Dholpur

  Intermediate; **Percentage: 66** — Apr 2021 – Mar 2022
- **National Public Sr. Sec. School** — Bari, Dholpur

  Matriculation; **Percentage: 74.17** — Apr 2019 – Mar 2020